ZHU ELLEN

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EDUCATION

IMPERIAL COLLEGE LONDON — LONDON, UK

Master of Science in Mathematics (MSci First Class Honor) — Sep 2022 – Sep 2023

Courses: Deep Learning, Data Science, Statistical Learning, Stochastic Differential Equations in Financial Modelling, Credit Risk Modelling, Stochastic Simulation, Classical Dynamics, Mathematical Biology

IMPERIAL COLLEGE LONDON — LONDON, UK

Bachelor of Science in Mathematics (First Class Honor) — Sep 2019 – Sep 2022

Courses: Machine Learning, Time Series, Option Pricing, Optimization, Scientific Computation, Statistical Modelling, Objected Oriented Programming, Statistical Modelling, Applied Probability, Calculus, PDE, Linear Algebra

WORK EXPERIENCE

MAKO TRADING — LONDON, UK

Graduate Trader — Sept 2025 – Present

- Researched STIR derivatives (SONIA, EURIBOR, SOFR futures and options), built a model to extract the implied probability distribution of STIR from futures pricing, decomposed market prices into meeting-by-meeting rate expectations using analytical approximations and identified mispricing between futures and implied policy paths
- Developed quantitative volatility and relative value strategies across futures and options markets, including variance decay analysis around MPC/ECB/FED meetings and delta-neutral strip trades exploiting mispricing in implied rate paths
- Analyzed Brazilian equity and index options markets using high-frequency trade and order book data, researching volatility surfaces, correlation dynamics, dispersion opportunities, and counterparty-dependent price impact and response functions
- Built flow-driven volatility forecasting and research frameworks in Python, incorporating net vega flow, gamma imbalance, aggressive flow reconstruction, and option structure classification to infer delta, gamma, vega, skew, and term-structure exposures across multi-leg options trading activity

BNP PARIBAS — LONDON, UK

FX Option Trader Internship — Jun 2025 – Sept 2025

- Developed a pricing model for USDHKD options incorporating a discrete jump-diffusion framework with up/down/stay scenarios and uniform distribution in the pegged band (7.75–7.85), enabling accurate pricing under pegged FX dynamics
- Built a PnL explainer tool to reconcile daily Present Value changes with theta and other risk factors, improving transparency of PnL attribution for trading desk
- Analysed intraday volatility contribution by constructing time-sliced volatility decomposition, providing insights into the distribution of market risk throughout the trading day

BARCLAYS BANK PLC — LONDON, UK

Quantitative Analytics — Aug 2023 – May 2025

- Cleaned financial time-series data by removing outliers, applied PCA transformation or Lasso to deal with multicollinearity
- Used Cross Validation to adjust the hyperparameter such as lags and assessed the model by p-values(usually smaller than 0.001) and out of sample R-squared values by Python
- Employed Monte Carlo simulation techniques to derive loss distributions such as trading error to gain Value at Risk
- Conducted data cleaning on the return time series on a global portfolio of 14 major asset classes, designed and implemented a growth factor mimicking portfolio, leveraging the constructed dataset, which explained 40% of GDP forecast deviations

CYCLONE ROBOTICS — SHANGHAI, CHINA

Machine Learning Internship — Jul 2022 - Sept 2022

- Analysed and interpreted visual data by K-Means clustering and seaborn, utilised deep learning frameworks such as PyTorch for the development, training, and optimisation of neural network models

RESEARCH PROJECTS & ACTIVITIES

PROXIMAL GRADIENT DESCENT ON CONVEX OPTIMIZATION — LONDON, UK

Master's Project — Jun 2022 – Aug 2023

- Compared with Sub-gradient Descent, Proximal Gradient Descent by Iterative Shrinkage Thresholding Algorithm, enhanced the efficiency of LASSO regression, a convex relaxation approach for solving all non-convex problems
- Utilised LASSO in Finite Element Method to get approximate solutions of Partial Differential Equations

WORLDQUANT BRAIN

GOLD LEVEL in WorldQuant Challenge — Sep 2023 – Present

- Analysed and leveraged over 100,000+ data fields from European and Asian markets to develop predictive Alpha model

SKILLS AND INTERESTS

- Languages and Platforms: Python, R, SQL, Microsoft Office Suites, LATEX, Git, Objected-Oriented Programming
- Interests: Climbing, Racing, Skating
- Prizes: A-level (A*A*A*) GCSE (Mathematics 1st in the world)